Competing Risks: Prepayment vs. Default

Part E(i) — Mortgage Survival Analysis

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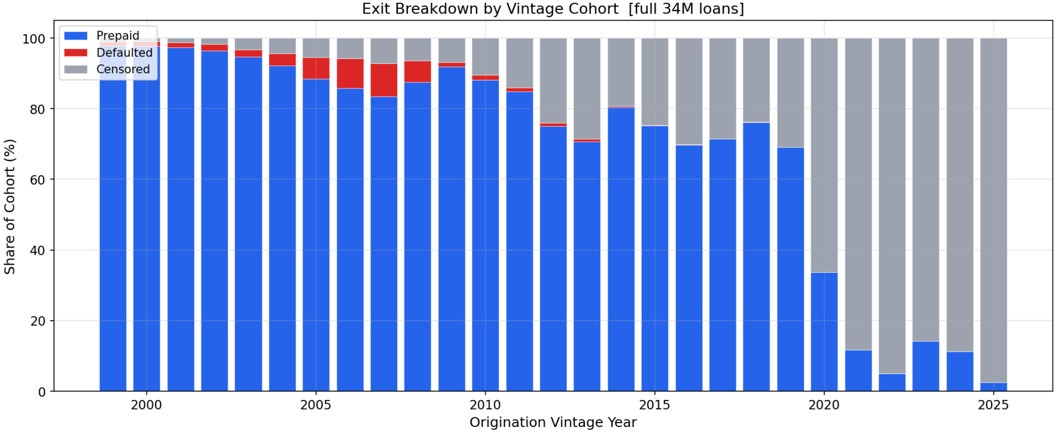
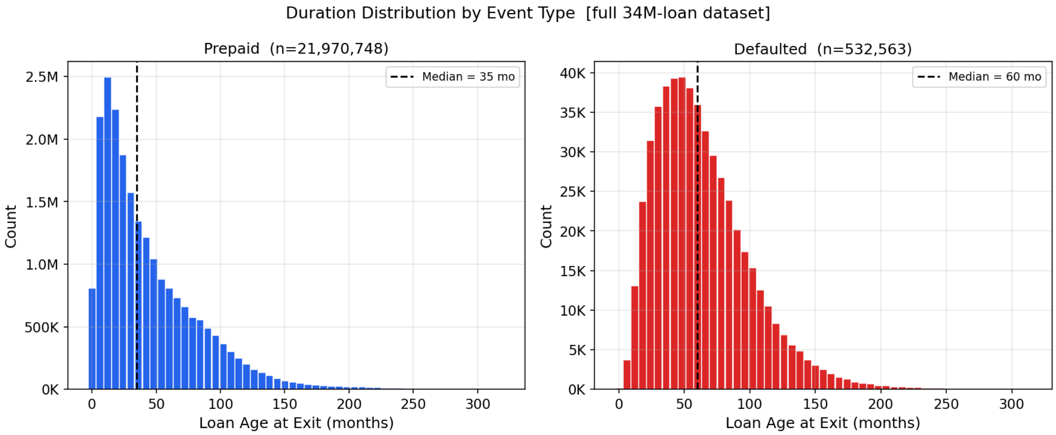
Vintage era — not FICO or LTV — is the dominant risk driver.



Two competing fates, two different borrower stories.



Two estimators, two different treatments of competing events.

Kaplan–Meier

$$\hat{F}^{(k)}_{\mathrm{KM}}(t) = 1 - \prod_{t_j \le t} \left(1 - \frac{d_j^{(k)}}{n_j^{\mathrm{KM}}} \right)$$

$d_j^{(k)}$	cause-k exits at t_j
n_j^{KM}	risk set excluding competing-event loans

Aalen–Johansen

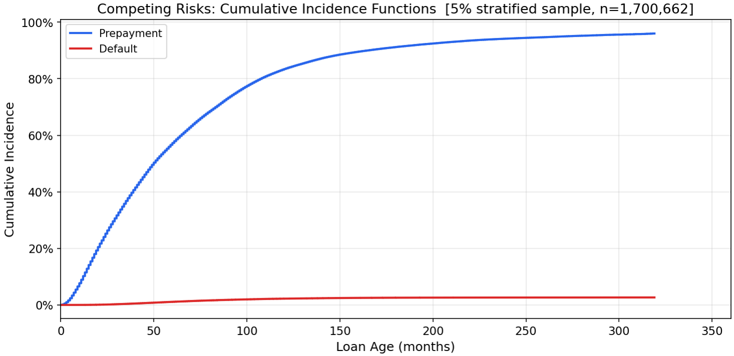
$$\hat{F}_k(t) = \sum_{t_j \le t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j}$$

d_{kj}	cause-k exits at t_j
n_j	risk set including all exits
$\hat{S}(t_j^-)$	overall survival just before t_j

KM: competing exits censored $\Rightarrow$ denominator too small $\Rightarrow$ each increment inflated.

AJ: $\hat{S}(t_j^-)$ shrinks with every exit $\Rightarrow$ $\hat{F}^{(\mathrm{pre})} + \hat{F}^{(\mathrm{def})} + \hat{S} = 1$ always.

The Aalen–Johansen estimator gives the only coherent probability partition.



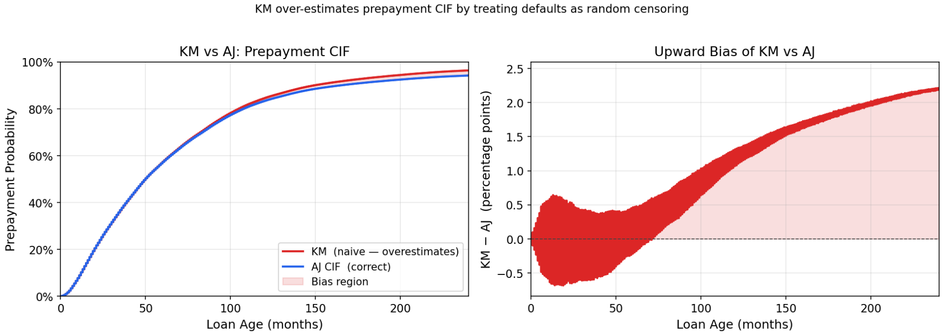
Cumulative Incidence Function

$$F_k(t) = P(\text{exit by } t,\ \text{cause } k)$$

Aalen–Johansen CIF partition

	Prepaid	Default	Active
5 yr	61.4%	1.1%	37.6%
10 yr	83.3%	2.2%	14.5%
20 yr	91.8%	3.1%	5.1%

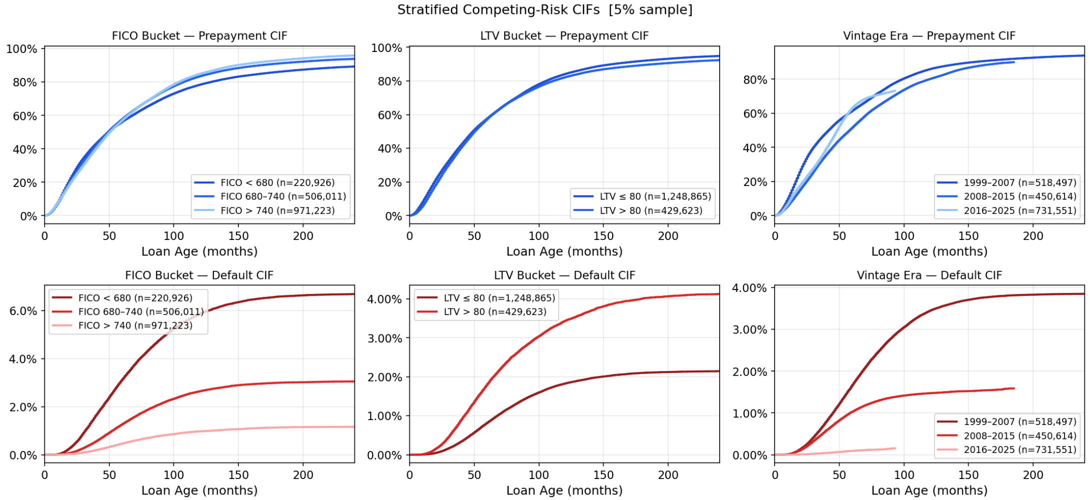
KM overstates prepayment probability because it ignores competing defaults.



KM overstatement of prepayment probability

Horizon	$\hat{F}_{\mathrm{KM}}$	$\hat{F}_{\mathrm{AJ}}$	Bias
5 yr	61.9%	61.4%	+0.45 pp
10 yr	84.6%	83.3%	+1.31 pp
20 yr	93.4%	91.8%	+2.23 pp

FICO separates default 6×; LTV separates default 2× — neither separates prepayment.



Cause-specific Cox: one separate hazard model per cause.

Cause-Specific Cox

$$h_k(t \mid \mathrm{x}) = h_{0k}(t)\, \exp\big(\boldsymbol{\beta}_k^\top \mathrm{x}\big), \quad k \in \{\mathrm{pre}, \mathrm{def}\}$$

$$\ell(\boldsymbol{\beta}_k) = \sum_{i:\, E_i = k} \left[\boldsymbol{\beta}_k^\top \mathrm{x}_i - \log \sum_{j \in \mathcal{R}(t_i)} e^{\boldsymbol{\beta}_k^\top \mathrm{x}_j} \right]$$

$\mathcal{R}(t_i)$: all loans still active just before t_i.

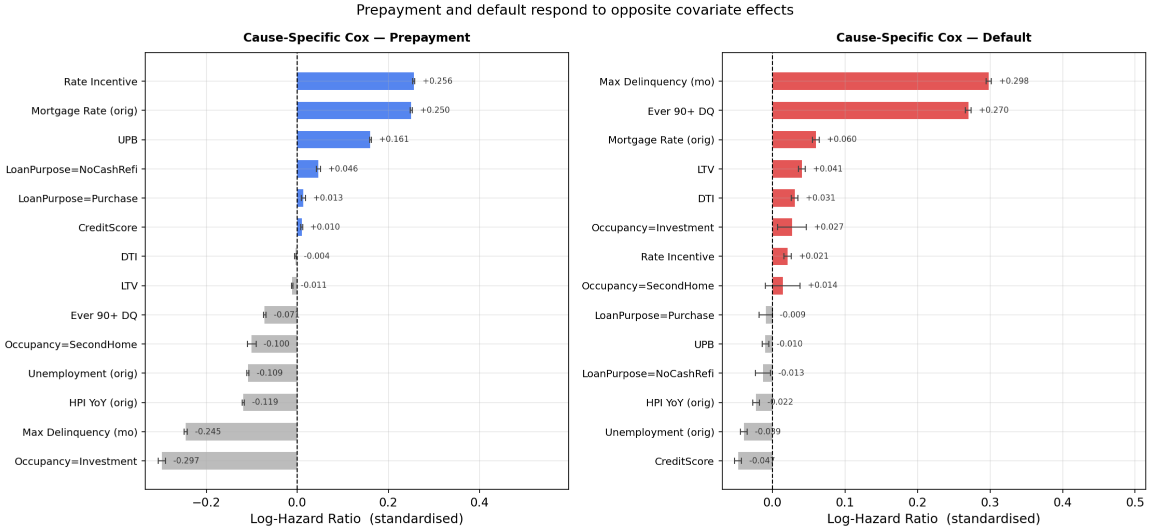
Competing-cause exits are **censored** — they leave $\mathcal{R}(t)$ without contributing to ℓ.

Additional features

Max Delinquency — months past due over loan life

Ever 90+ DQ — binary flag for severe delinquency

Prepayment and default respond to opposite covariate effects.



Four covariates flip sign — a combined model cancels both signals.

Naive Cox vs. cause-specific coefficients

Covariate	**Naive $\hat{\beta}$**	**Prepay $\hat{\beta}$**	**Default $\hat{\beta}$**	
Rate incentive	$+0.253$	$+0.256$	$+0.021$	(pure prepay)
UPB	$+0.161$	$+0.161$	-0.010	(pure prepay)
FICO	$+0.016$	$+0.010$	-0.047	**opposite**
LTV	-0.007	-0.011	$+0.041$	**opposite**
Max Delinquency	-0.123	-0.245	$+0.298$	**opposite**
Ever 90+ DQ	-0.035	-0.071	$+0.270$	**opposite**

Each extra month delinquent raises default hazard by 35% and cuts prepayment hazard by 22% — yet the naive coefficient (-0.123) conceals both, collapsing a $+0.543$ spread to near zero.

Key Takeaways

1. **Default and prepayment are competing risks**
   A naive Cox model is biased and conflates two economically opposite processes.
2. **Kaplan–Meier overstates prepayment probability by up to $+2.2$ pp**
   It ignores the competing cause absorbing risk; Aalen–Johansen is required.
3. **CIF curves reveal large cohort-level variation**
   The 2007 cohort's default CIF is $6\times$ the long-run average; post-2020 prepayment CIF is nearly flat due to rate lock-in.
4. **FICO drives default; rate incentive drives prepayment**
   Each feature is essentially invisible to the other cause — the two risks operate through separate channels.
5. **Fine–Gray and cause-specific Cox give similar results here**
   The gap opens when default and prepayment are strongly correlated — crisis vintages or high-risk borrowers.

Thank you

Questions?

Fine & Gray (1999) *JASA* · Aalen & Johansen (1978) *Scand. J. Stat.* · Sadhwani et al. (2021) *J. Fin. Econometrics*